

ALAN WANG

🏠 [urd00m.github.io/](https://github.com/urd00m) ✉ wangalan@berkeley.edu 🌐 [urd00m](https://github.com/urd00m)
in [alan-wang-urd00m/](https://github.com/alan-wang-urd00m/) 🎓 Google Scholar 📍 Berkeley, CA, USA

RESEARCH INTERESTS

I have broad interests in side-channel attacks and defenses, AI for hardware security, computer architecture, security, and systems. In the past, I have also been interested in drug discovery and MD simulations.

EDUCATION

University of California, Berkeley
Ph.D. in Computer Science

Berkeley, California
2025–2030

University of Illinois at Urbana-Champaign
Conjoined BSMS in Computer Science

Urbana, Illinois
2021–2025

PUBLICATIONS

- [1] First author. “Redacted”. In: *Redacted*. 2026.
- [2] “Redacted”. In: *Redacted*. 2026.
- [3] Alan Wang et al. “Pixnapping: Bringing Pixel Stealing out of the Stone Age”. In: *Proceedings of the 32nd ACM SIGSAC Conference on Computer and Communications Security (CCS)*. 2025. URL: <https://www.pixnapping.com>.
- [4] Siyuan Chai et al. “EMT: An OS Framework for New Memory Translation Architectures”. In: *Proceedings of the USENIX Symposium on Operating Systems Design and Implementation (OSDI)*. 2025.
- [5] Alan Wang et al. “Peek-a-Walk: Leaking Secrets via Page Walk Side Channels”. In: *Proceedings of the IEEE Symposium on Security and Privacy (S&P)*. 2025.
- [6] Rutvik Choudhary et al. “Declassiflow: A Static Analysis for Modeling Non-Speculative Knowledge to Relax Speculative Execution Security Measures”. In: *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security (CCS)*. 2023.
- [7] Alec Auster et al. “MITHRIL: Mars Ice Thermal Harvesting Rig and ISRU Laboratory”. In: *Proceedings of AIAA ASCEND*. 2022.

HONORS & AWARDS

- NDSEG Fellowship, *Department of Defense* 2026
- Apple Bug Bounty Award, *Apple* 2026
- IEEE Micro Top Picks Honorable Mention, *IEEE* 2026
- Jane Street GRF Finalist, *Jane Street* 2026
- 3x High-Severity Bug Bounty Awards, *Google* 2025
- NSF GRFP Honorable Mention, *National Science Foundation* 2025
- Bronze Tablet Scholar Award, *University of Illinois at Urbana-Champaign* 2025
- Siebel Scholar Award, *Siebel Scholars Foundation* 2025
- Best Poster in Theme, *SRC PRISM* 2025

- Dean's List, *Grainger College of Engineering, UIUC* 2021–2023
- 2nd Place Overall, *NASA RASC-AL Competition* 2022

BUGS AND NEWS

CVEs: CVE-2025-48561, CVE-2025-48630, CVE-2026-28964

Highlighted news: [Ars Technica](#), [Malwarebytes](#), [Risky Business](#), [LWN](#)

EXPERIENCE

University of California, Berkeley

Graduate Student Researcher

Berkeley, CA, USA

August 2025 – Current

- Redacted: two works under submission
- Recent work: [pixnapping.com](#). This work presents a new class of side-channel attacks on Android devices that can steal graphically rendered secrets. Most notably, we demonstrate leaking ephemeral app-based 2FA codes in under 30 seconds
- Attacks: researching side-channel attacks inherent to graphics stacks (i.e., pixel stealing attacks)
- Defenses: exploring defenses for the side-channel vulnerabilities found by my attack research

Jane Street

Software Developer Intern

New York, NY, USA

May – August 2024

- Worked with billions of trade orders submitted to the SEC
- Improved the efficiency and utilization of servers used by the entire firm

D. E. Shaw Research

Computer Science Intern

New York, NY, USA

May – August 2023

- Researched docking and non-equilibrium FE calculation methods
- Wrote system software which ran thousands of simulations on Anton3 ASICs and achieved a 2.5x performance improvement

FPSG Lab

Research Assistant

Urbana, IL, USA

September 2021 – May 2025

- Created a new class of side-channel attacks on Android devices.
- Created a new general-purpose cache side channel exploiting page walkers and used it to exploit the Linux kernel using Spectre-V2 and the Intel DDP
- Helped build a new Linux framework to support new, radical memory translation architectures
- Modelled non-speculative information flow to decrease mitigation overhead by up to 21%

Argonne National Lab

Research Aide

Lemont, IL, USA

May 2022 – May 2023

- Worked with NVIDIA's Bluefield-3 Data Processing Unit (DPU) for zero trust network architectures
- Found critical errors by instrumenting Portable Batch System (PBS) for Argonne's extreme scale systems
- Developed a command line interface for Argonne's UserBase3, used by all Argonne admins

Argonne National Lab

Visiting Student

Lemont, IL, USA

February – May 2022

- Led the design of a ROS2 interface for Argonne's self-driving lab
- Built key infrastructure for Argonne's self-driving lab

Northwestern University
Undergraduate Research Assistant

Evanston, IL, USA
February 2021 – June 2022

- Researched the vulnerability INTEL-SA-00086 to gain access to Intel’s most secure piece of hardware (microcode project)

Argonne National Lab
DoE College Bound Research Intern (CBRP)

Lemont, IL, USA
June – August 2021

- Started the design of a ROS2 interface for Argonne’s self-driving lab
- Created important building blocks for future work in Argonne’s self-driving lab

Office of Naval Research
SEAP Intern

Crane, IN, USA
Summers 2019 & 2020

- Led the development of autonomous bomb-defusing robots
- Repaired software related to navigation, object recognition, and arm manipulation

MENTORING

- Jonathan Ji, *Undergrad Research Mentor* 2026–current
- Nathan Vaz, *Undergrad Research Mentor* 2026–current
- Matin Sadeghian, *Undergrad Research Mentor* 2026–current
- Steven Tang, *Undergrad Research Mentor* 2026–current
- Zaheen Jamil, *Undergrad Research Mentor* 2026–current
- Ailsa Sun, *Undergrad Research Mentor* 2026–current
- Kaustubh Khulbe, *Undergrad Research Mentor* 2025 spring
- Ishan Buyyanapragada, *ACM Mentor* 2024–2025
- Spencer Acquah, *ACM Mentor* 2023–2024
- Christopher Suarez, *ACM Mentor* 2023–2024

SERVICE

- Berkeley Security Seminar cohost, 2026–current
- ACM mentor, 2022–2025
- UIUC course grader, 2023
- Self-organized competitive programming course instructor to 30 K–12 students, 2021
- Self-organized Java course instructor to over 100 K–12 students, 2020

TALKS & PRESENTATIONS

- Redacted, *Intel Labs* August 2025
- Redacted, *Apple* August 2026
- Redacted, *Intel Labs* July 2026
- Pixnapping: Bringing Pixel Stealing out of the Stone Age, *Berkeley Security Seminar* February 2025
- Pixnapping: Bringing Pixel Stealing out of the Stone Age, *Stanford Security Seminar* February 2025
- Pixnapping: Bringing Pixel Stealing out of the Stone Age, *SLICE Retreat* January 2025
- Pixnapping: Bringing Pixel Stealing out of the Stone Age, *ACM CCS* October 2025
- Peek-a-Walk: Leaking Secrets via Page Walk Side Channels, *Intel IPAS* July 2025
- Peek-a-Walk: Leaking Secrets via Page Walk Side Channels, *IEEE S&P (Oakland)* May 2025

TECHNICAL SKILLS

Programming Languages: C, C++, Python, x86 & ARM Assembly, OCaml, Bash, Rust, Java, Swift, SystemVerilog, Fortran, C#

Systems & Tools: Linux, ROS/ROS2, Git, L^AT_EX, LLVM, Android, iOS, macOS, Windows